Vyora

Project Management Brief & System Architecture Documentation

Version: 1.4.0 (Enterprise Ready)
Date: October 2026
Environment: Hybrid Multi-Tenant Monolith

1. Executive Summary & Product Architecture

Vyora is a production-grade multi-tenant GST billing, receivables, supplier payables, and traceable inventory management platform for Indian small-to-medium businesses (retailers, wholesalers, manufacturing, and distributors).

vyora-backend

NESTJS 11 + MONGOOSE 8

Decoupled modular REST service. Multi-tenant boundary isolation enforced via `TenantGuard`, `RolesGuard`, and `PlanGuard`. Minor currency integer precision across all ledgers.

vyora-frontend

REACT 19 + VITE 6 + TAILWIND 4

Mobile-first administrative dashboard using Theme B — Teal Ledger design system. Dynamic feature gating, responsive tables with horizontal scroll safeguards, and real-time role switchboard.

vyora-website

NEXT.JS MARKETING & BLOG

Public marketing site at `D:\project\vyora-website` featuring product capabilities, pricing plans (Starter, Growth, Business Pro), and SEO content blog.

vyora-mobile

PLANNED REACT NATIVE

Upcoming cross-platform mobile POS & counter invoicing app consuming the existing authenticated backend API.

2. Role-Based Access Control (RBAC) System

The platform strictly separates authority between business owners and operational staff to protect financials, catalog integrity, and audit compliance.

USER ROLE	PERMISSION LEVEL	PERMITTED CAPABILITIES	RESTRICTIONS & SAFEGUARDS
OWNER / ADMIN	FULL MANAGEMENT CONTROL	• Create, Edit, Finalize, and Cancel Invoices • Add and Archive Products & Catalog Items • Add and Edit Customers & Suppliers • Record Purchases & Payment Vouchers • Adjust Stock (Business Pro Plan) • Switch user roles and toggle active status • Upgrade or change subscription plans	• Cannot demote or suspend the last remaining owner in the business. • Audit trail logs identity for all modifications.
STAFF	VIEW-ONLY OPERATIONAL	• View Dashboard revenue metrics & stats • View & Search Invoices, preview & print PDFs • Browse Product catalog, units & prices • Browse Customer & Supplier ledgers	• **Zero mutation rights:** All Add, Edit, Delete, Finalize, Cancel buttons hidden. • Backend `RolesGuard` rejects mutation requests with 403 Forbidden.

USER ROLE	PERMISSION LEVEL	PERMITTED CAPABILITIES	RESTRICTIONS & SAFEGUARDS
		• Inspect stock levels and movement logs	

3. Dedicated Team & Roles Management (`/roles`)

A centralized management screen allowing business owners to monitor staff accounts, invite team members, and toggle roles instantly with live database synchronization.

- **Interactive Role Switch:** Each team member has a dedicated switch toggle. Switching ON grants `OWNER (Full Access)`; switching OFF sets `STAFF (View Only)`. Changes apply immediately via `PATCH /business/staff/:id/role`.
- **Access Status Toggle:** An active status switch allows owners to temporarily suspend or re-activate member access via `PATCH /business/staff/:id/status` without deleting their records.
- **Protective Guardrails:** The last remaining owner is protected by backend invariants preventing demotion or suspension, ensuring the business is never locked out.
- **Member Invitation Modal:** Owners can invite new staff members with required fields marked with prominent red asterisks (*). System sets standard initial credentials for frictionless onboarding.

4. Subscription Tiers & Feature Gating Matrix

Features throughout the application are gated based on the active `subscriptionPlan` on the `Business` record, aligning exactly with the public marketing website:

FEATURE MODULE	STARTER (₹0 / FREE)	GROWTH (₹499 / MO)	BUSINESS PRO (₹999 / MO)
GST Invoicing	INCLUDED (50 / MO)	UNLIMITED	UNLIMITED
Product Catalog & Inventory	UP TO 100 PRODUCTS	UNLIMITED + STOCK ALERTS	UNLIMITED + STOCK ALERTS
Customer Directory & Dues	BASIC BALANCES	CREDIT LIMITS & LEDGERS	CREDIT LIMITS & LEDGERS
Supplier Purchases & Payables	HIDDEN (UPGRADE)	FULL INBOUND BILLS	FULL INBOUND BILLS
Payments & Receipts Vouchers	HIDDEN (UPGRADE)	FULL CASH/BANK VOUCHERS	FULL CASH/BANK VOUCHERS
Tax Reports (GSTR-1, 3B, Valuation)	HIDDEN (UPGRADE)	FULL TAX REGISTERS	FULL TAX REGISTERS
Team Staff Delegation (`/roles`)	1 SOLO OWNER ONLY	UP TO 3 STAFF ACCOUNTS	UNLIMITED STAFF ACCOUNTS
Audited Stock Adjustments	HIDDEN	HIDDEN	FULL AUDITED ADJUSTMENTS
Financial Audit Trail Logs	STANDARD LOGS	STANDARD LOGS	COMPLETE AUDIT TRAIL

5. High-Concurrency Platform Stability & Enterprise UX (Scale: 1k - 50k Users)

To ensure flawless operation under concurrent load of 1,000 to 50,000+ active tenant users, comprehensive rate limiting, query throttling, table pagination, and security controls were integrated:

- **Sliding-Window Rate Limiting (`rate-limit.middleware.ts`)**: In-memory sliding-window limiter with a periodic 10-minute sweep to prevent memory leaks under 50k unique IP addresses. Restricts authentication routes (`/auth/*`) to 25 requests/minute (brute-force defense) and general API endpoints to 200 requests/minute. Injects standard `X-RateLimit-*`, `Retry-After`, and defense headers (`X-Content-Type-Options: nosniff`, `X-Frame-Options: SAMEORIGIN`, `X-XSS-Protection: 1; mode=block`).
- **Client Search Debouncing (`SearchInput.tsx`)**: 300ms internal keystroke debounce across product, invoice, party, and team queries. Prevents request storms, DOM thrashing, and unnecessary backend database indexing overhead while users type.
- **High-Capacity Data Table Architecture & Pagination**: Dedicated `<Pagination>` component deployed across `ProductTable`, `PartyTable`, `InvoiceTable`, and `StockMovementTable`. Supports dynamic page ranges, ellipsis navigation, entry counts, and configurable page limits (10, 25, 50, 100). Prevents out-of-memory (OOM) browser freezing when businesses manage thousands of ledger items.
- **Clean Icon-Only Action Buttons**: Table action columns feature minimalist, high-contrast icon-only buttons (`<Edit2 />` and `<Trash2 />`) with subtle hover pills and tooltips, eliminating button text clutter.

- **Prohibition of "Archive" Terminology**: In accordance with core requirements, ambiguous "Archive" labeling and functionality has been completely removed across UI and prompts in favor of explicit, unambiguous "Delete" and "Remove".
- **Modern SaaS Modal System & Navbar Cleanliness**: Modals upgraded with frosted backdrops (`bg-slate-900/60 backdrop-blur-sm`), smooth `zoom-in-95` entry animations, and responsive mobile stacks. Top navigation bar cleaned up by removing redundant role badges.
- **Authentic Plan Purchase & Checkout Flow (`PlanPurchaseModal.tsx`)**: Replaces direct 1-click plan switching with an authentic checkout experience: Monthly vs Annual (20% discount) billing cycle toggle, strict minor-unit paisa calculation with itemized 18% GST preview, payment method selection (UPI, Credit/Debit Cards, Net Banking), and simulated banking gateway verification with printable transaction receipts.

6. Engineering Milestones (V1-01 through V1-24)

ID	FEATURE MODULE	STATUS	IMPLEMENTATION DETAILS
V1-01	Auth & Multi-Tenancy	COMPLETED	JWT Bearer authentication, tenant header (`x-business-id`), membership isolation.
V1-02	Party Directory (CRM)	COMPLETED	Customers & Suppliers with GSTIN, 2-digit state codes, running balance ledger.
V1-03	Product Catalog	COMPLETED	SKU, HSN code, units, minor pricing (paisa), statutory GST tax rates (0-28%).
V1-04	GST Tax Calculations	COMPLETED	Intra-state (CGST+SGST) vs Inter-state (IGST) split, zero floating-point rounding error.
V1-05	Invoices & Sequence	COMPLETED	Immutable draft/finalized sequence numbers, atomic stock deductions, PDF generation.
V1-06	Traceable Inventory	COMPLETED	Audit log ledger for sales, purchases, cancellations, and manual adjustments.
V1-07	Purchases & Bills	COMPLETED	Inbound supplier purchase entry, inventory replenishment, and accounts payable.
V1-08	Payments & Vouchers	COMPLETED	Idempotent receipts and payment vouchers with automatic party balance reconciliation.
V1-09	Core Tax Reports	COMPLETED	Sales Register, Purchase Register, Stock Valuation, and Customer Debtor Aging.
V1-10	Business Settings	COMPLETED	Custom invoice prefixing, digit padding, legal terms, and audit log inspection.
V1-11	Database Persistence	COMPLETED	Full live connection to MongoDB Atlas Cloud cluster; static mock data eliminated.
V1-12	Fallback Empty States	COMPLETED	Contextual `EmptyState` components across all tables and charts.
V1-13	Form Validation & Badges	COMPLETED	Mandatory fields highlighted with red asterisks (*); Zod schema validations.
V1-14	Theme B: Teal Ledger	COMPLETED	Deep teal palette (`#0F766E`, `#123B3A`, `#F4F8F7`) across all components.

ID	FEATURE MODULE	STATUS	IMPLEMENTATION DETAILS
V1-15	Mobile Responsiveness	COMPLETED	Fixed thumb navigation bar, table horizontal scrolling safeguards, stacked buttons.
V1-16	Marketing Website	COMPLETED	Live public site at port 3001 with interactive pricing table and SEO blog.
V1-17	Automated AI Skill	COMPLETED	`vyora-core` skill providing architectural blueprint and engineering memory.
V1-18	Role Authorization (RBAC)	COMPLETED	Owner full access vs Staff view-only access enforced on both backend and frontend.
V1-19	Roles Switchboard Page	COMPLETED	Dedicated `/roles` page with individual role switches and access toggles.
V1-20	Subscription Plan Gating	COMPLETED	Starter, Growth, and Business Pro tier enforcement with dynamic feature hiding.
V1-21	Rate Limiting & Stability	COMPLETED	Sliding-window memory rate limiter + security headers for 1,000 to 50,000+ users.
V1-22	Search Input Debounce	COMPLETED	300ms internal debouncing across all search inputs to prevent server query spam.
V1-23	Table Pagination & Actions	COMPLETED	Pagination component, icon-only edit/delete actions, complete removal of archive.
V1-24	Plan Purchase Checkout	COMPLETED	Multi-step checkout modal with cycle toggles, 18% GST calculation, and UPI/Cards.